

Immune system basics

How tumors 'hide'

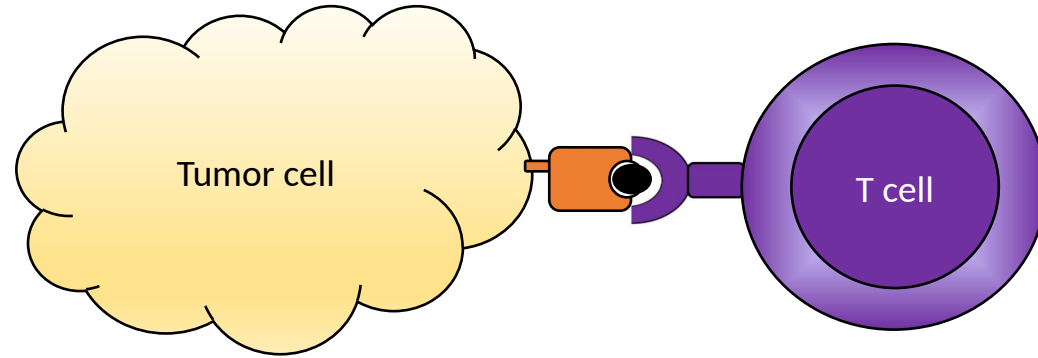
Immune checkpoints

Immune checkpoint
inhibitors (ICIs)

When do we use ICIs?

Side effects of ICIs

CASES



Immune Checkpoint Inhibitors

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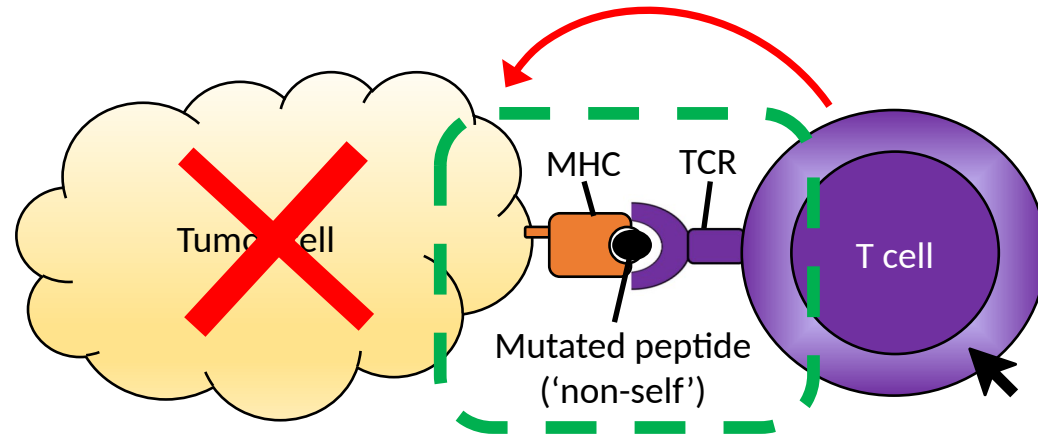
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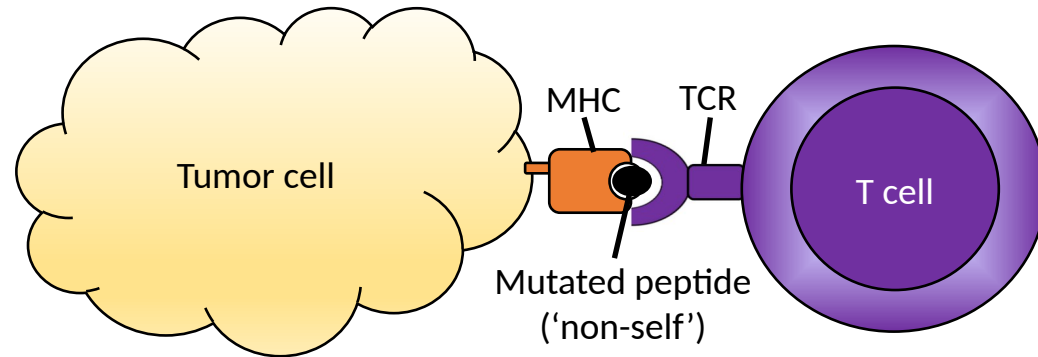
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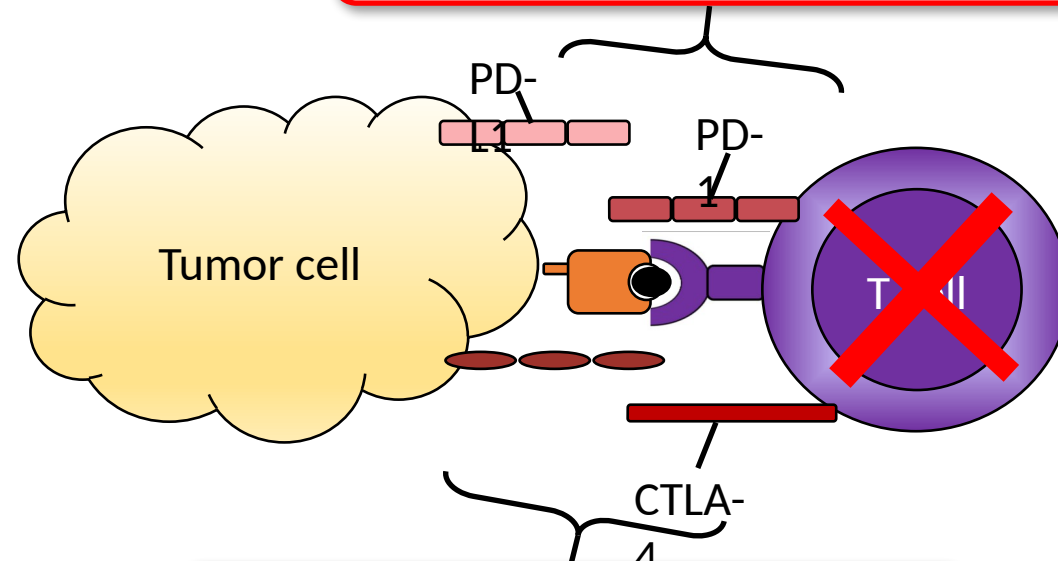
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PD-L1 and PD-1 interact and induces
T cell 'exhaustion'



CTLA-4 interacts with several ligands
and induces T cell 'exhaustion'

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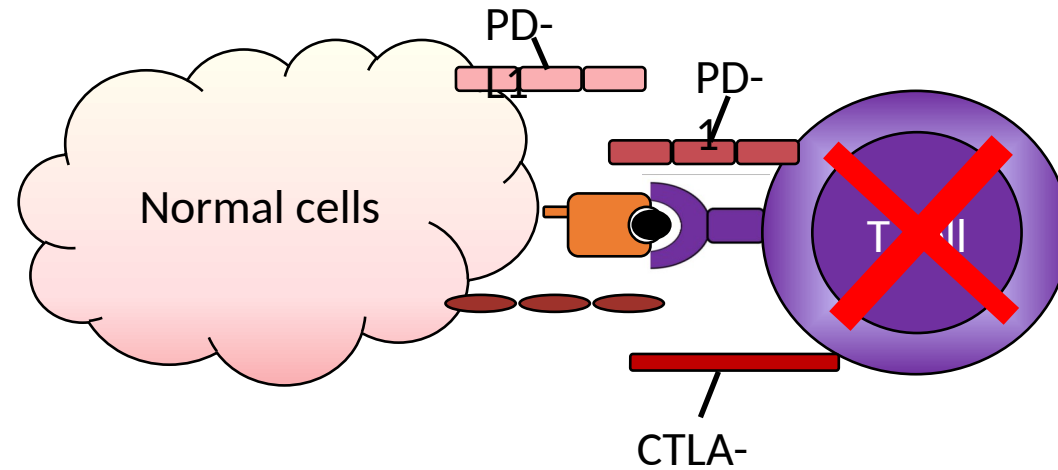
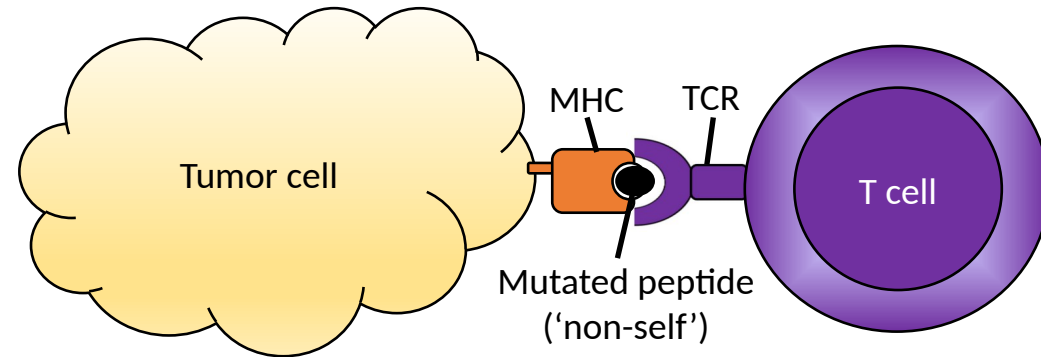
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Immune checkpoints are mechanisms by which
our immune system prevent autoimmunity

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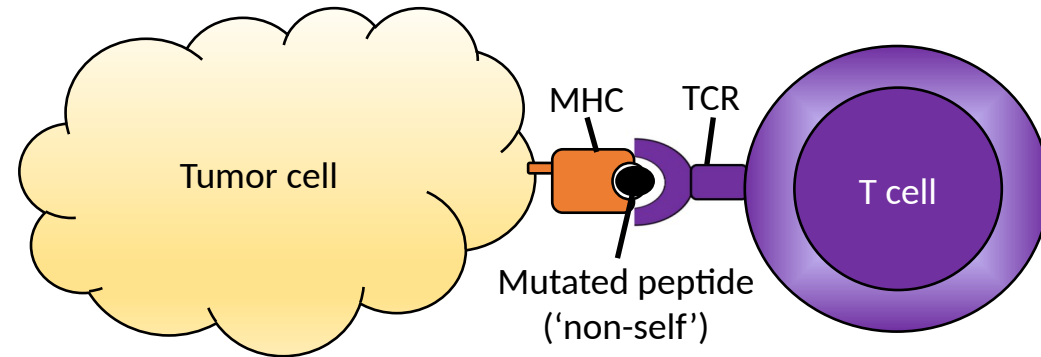
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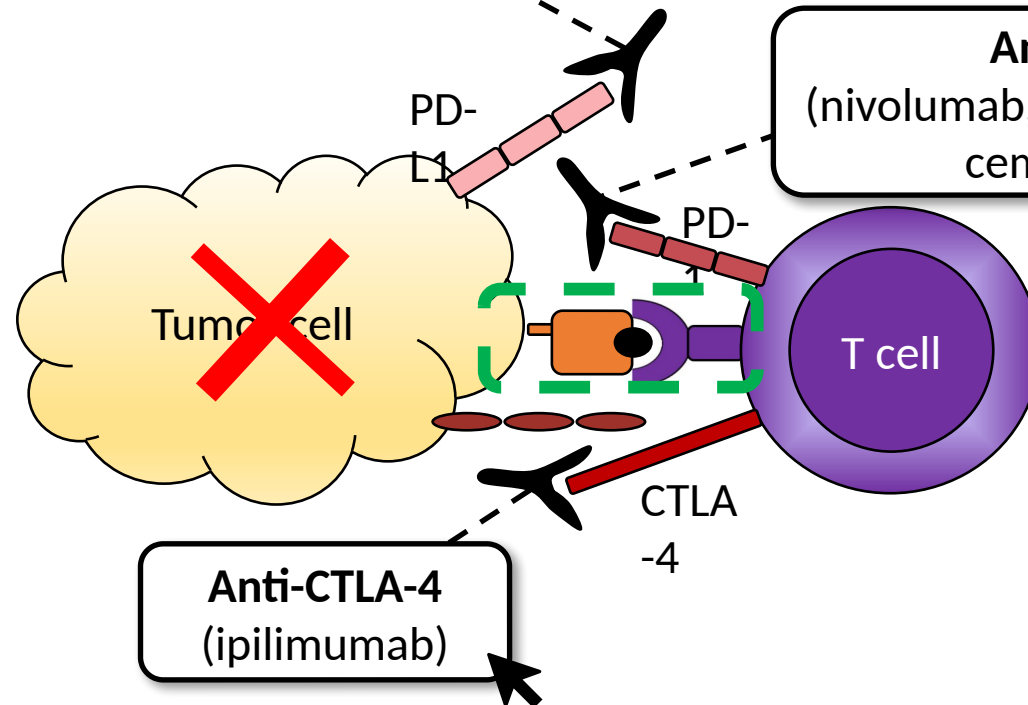
CASES



Anti-PD-L1
(avelumab, atezolizumab, durvalumab)

Anti-PD-1
(nivolumab, pembrolizumab, cemiplimab)

Anti-CTLA-4
(ipilimumab)



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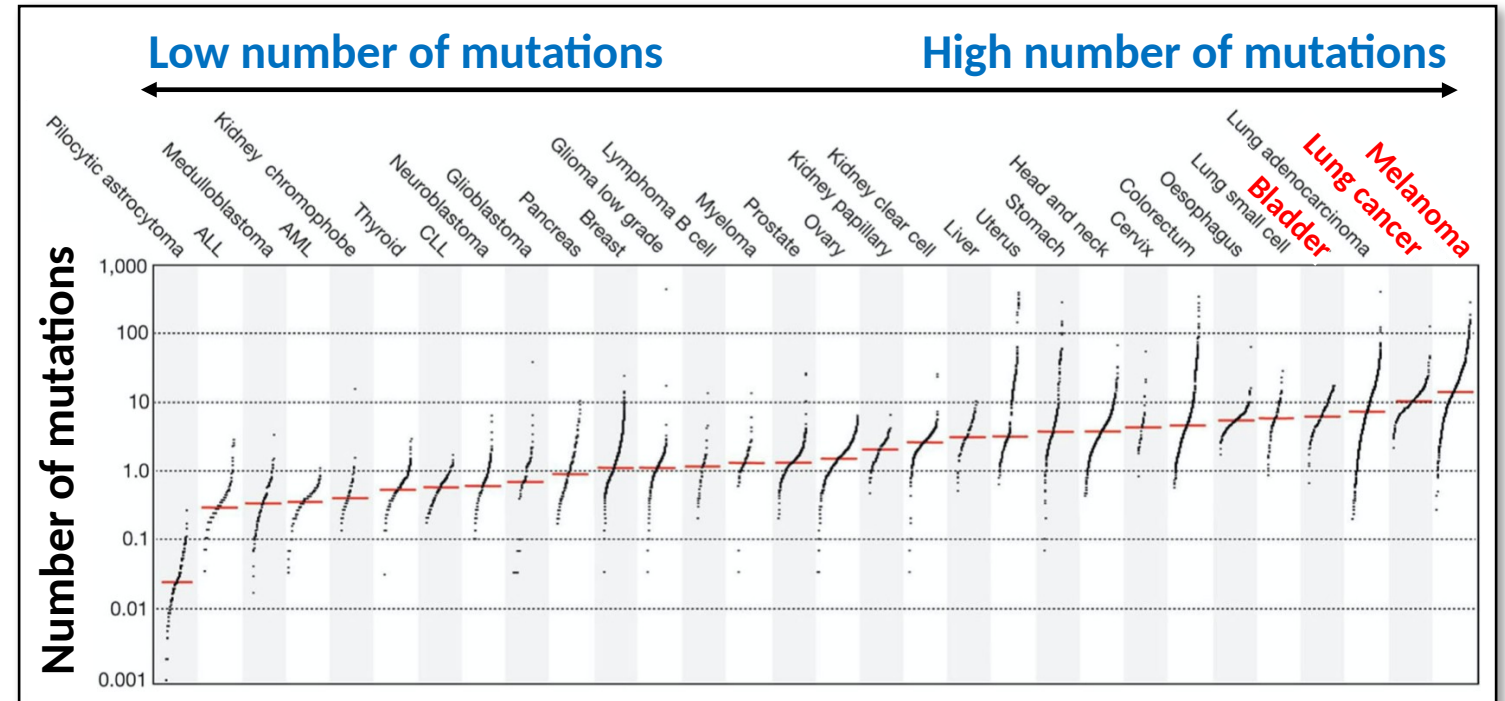
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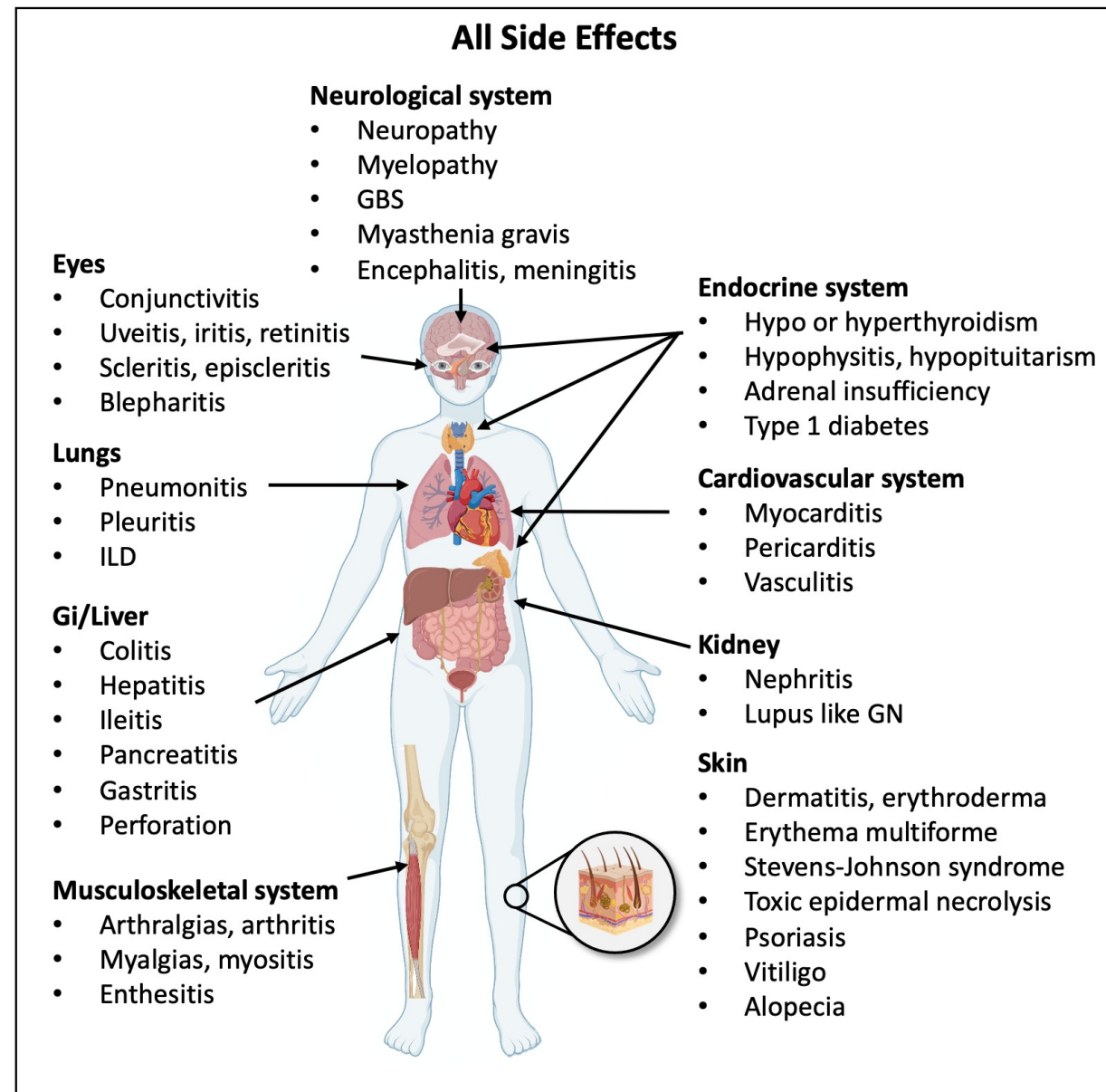
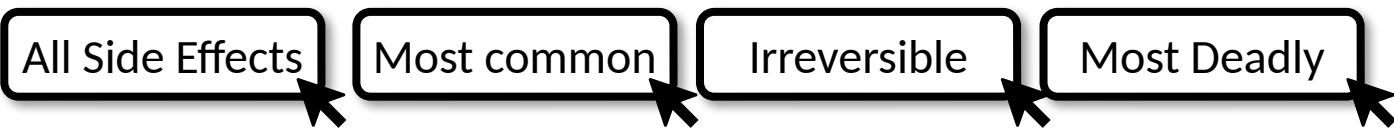
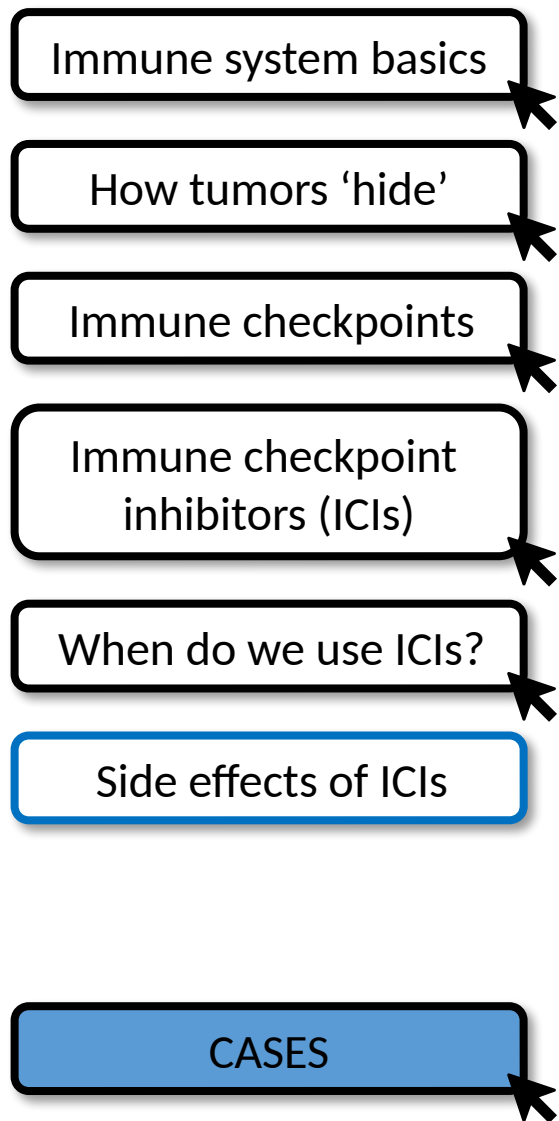


Which cancers have a lot of mutations? ²

Tumors that developed from recurrent exposure to mutagens such as melanoma (UV light), lung cancer (smoking)

Which cancers have immunotherapy FDA approval?

Head & neck cancer, melanoma, Merkel cell carcinoma, hepatocellular carcinoma, renal cell carcinoma, endometrial carcinoma, cervical cancer, lung cancer (small cell and non-small cell), malignant pleural mesothelioma, triple negative breast cancer, gastric & GEJ cancer, classical Hodgkin's lymphoma, urothelial carcinoma (BCG-unresponsive).



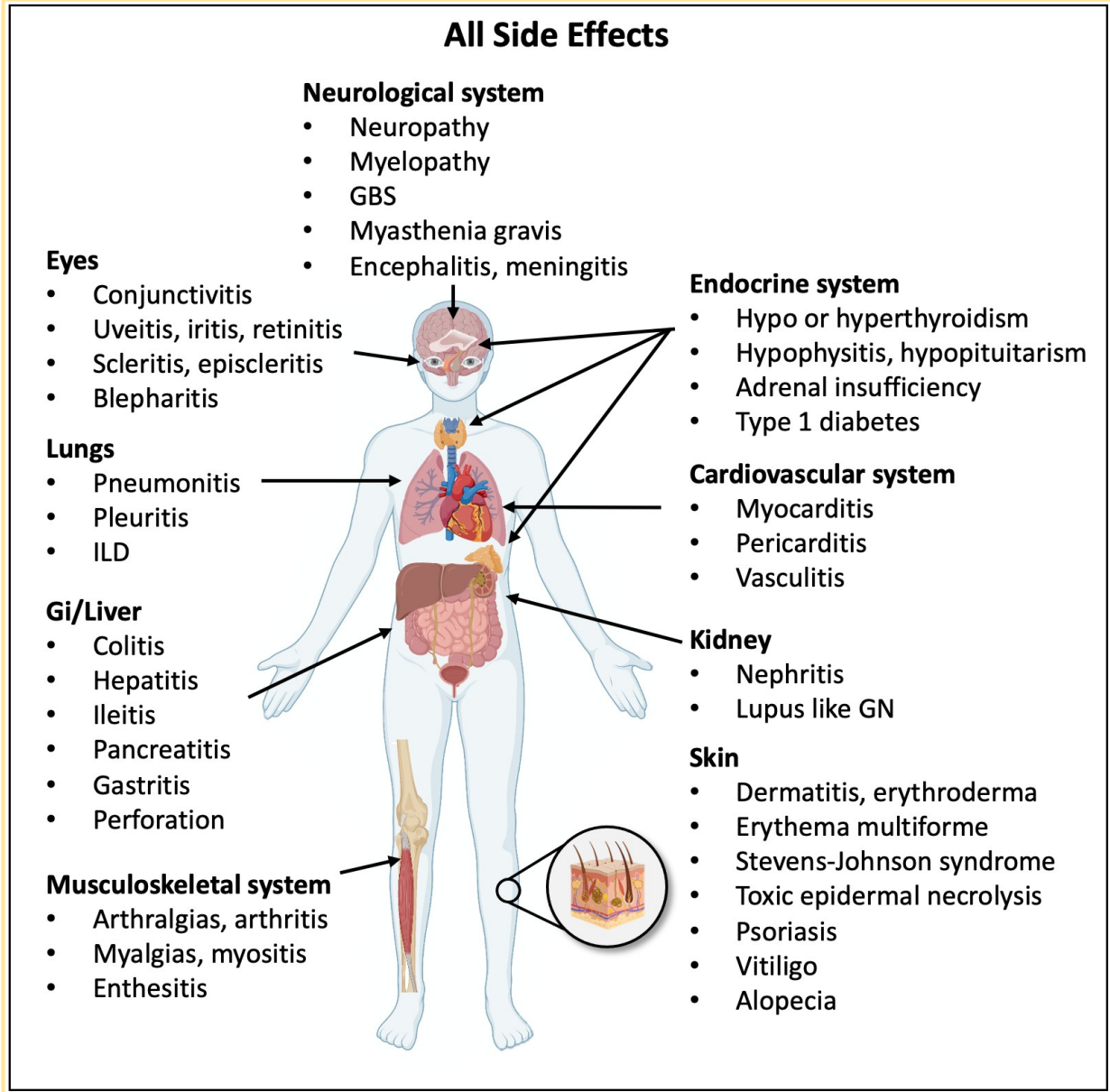
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All Side Effects

Most common

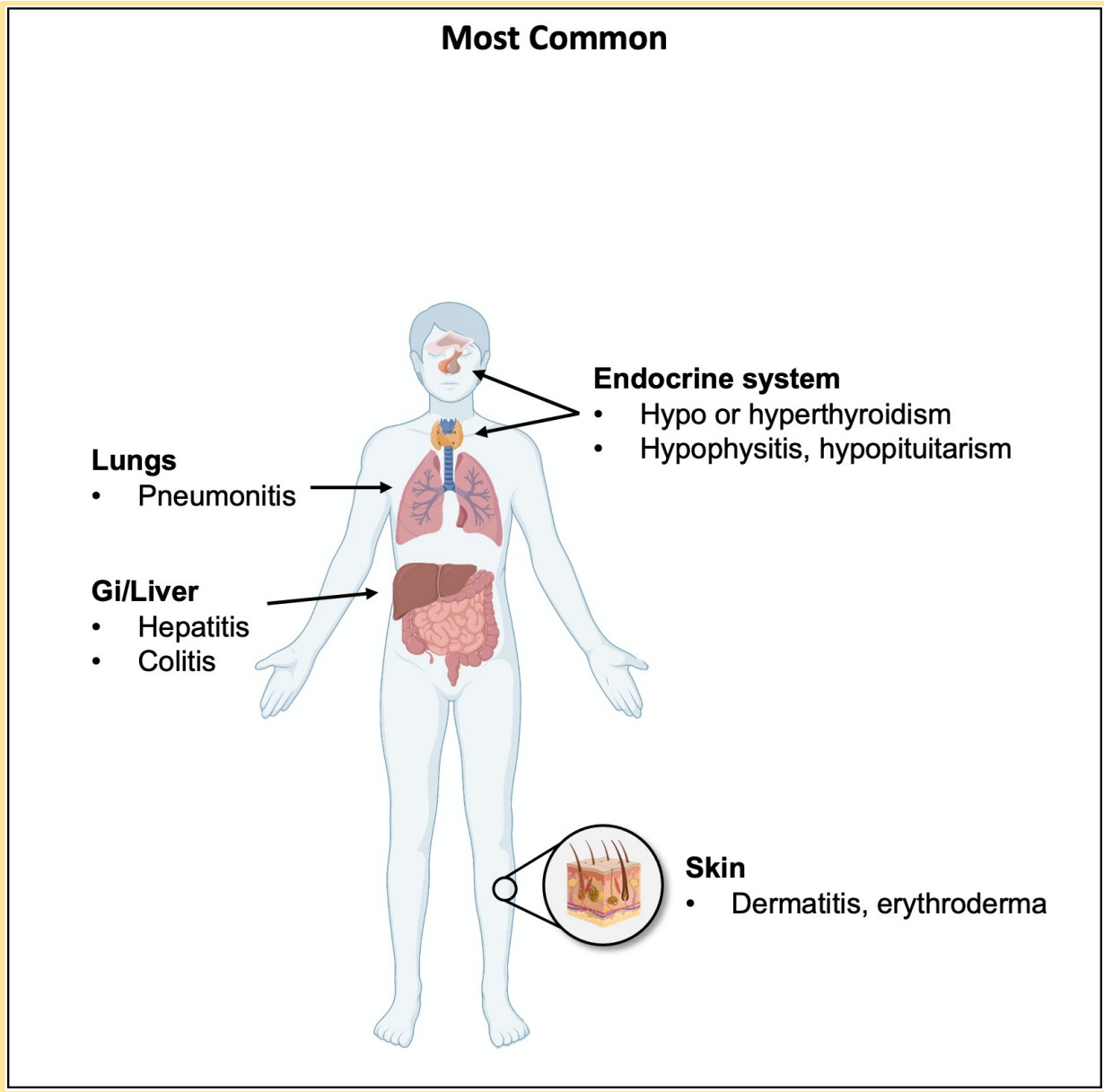
Irreversible

Most Deadly



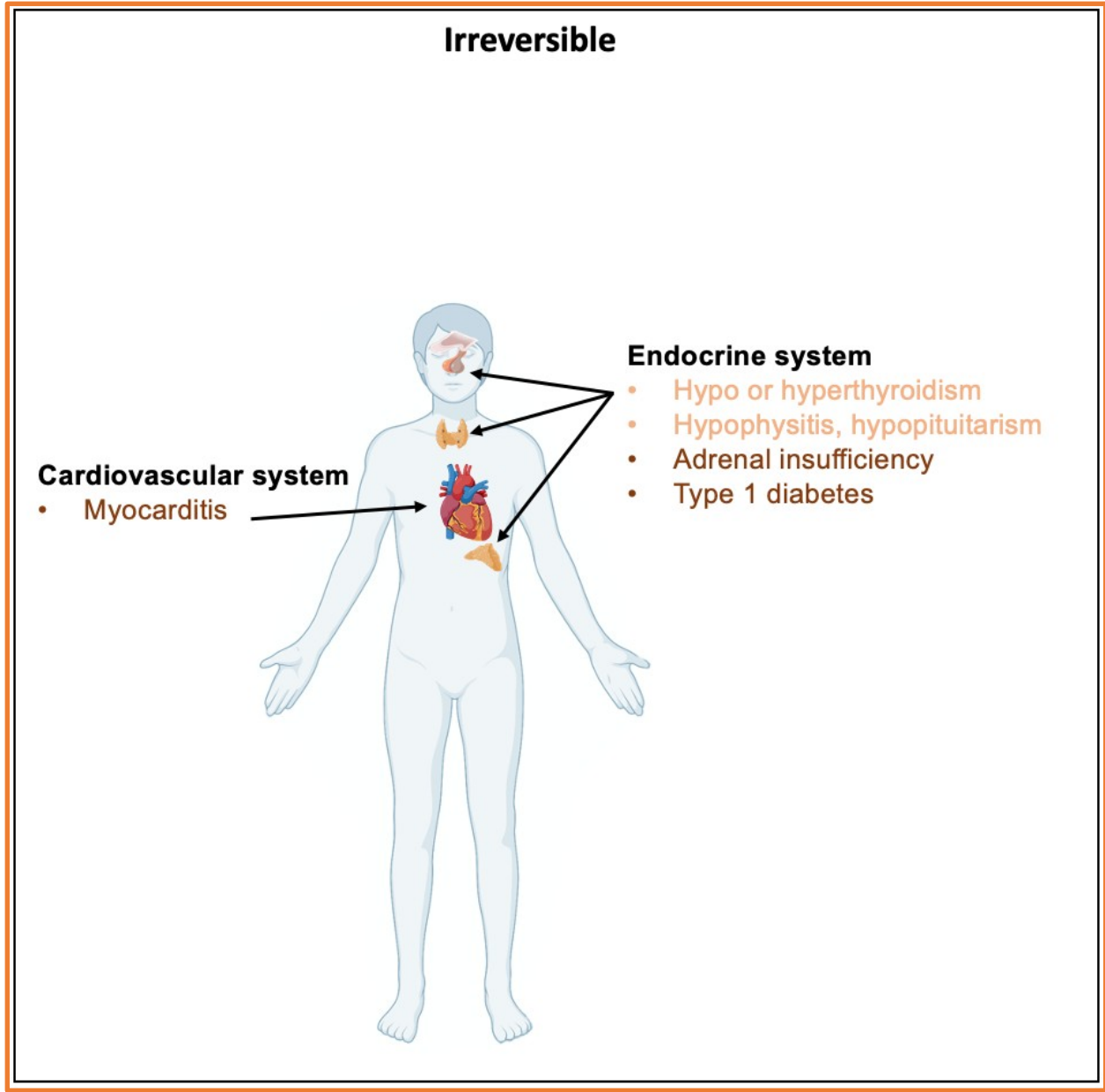
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- All Side Effects
- Most common
- Irreversible
- Most Deadly



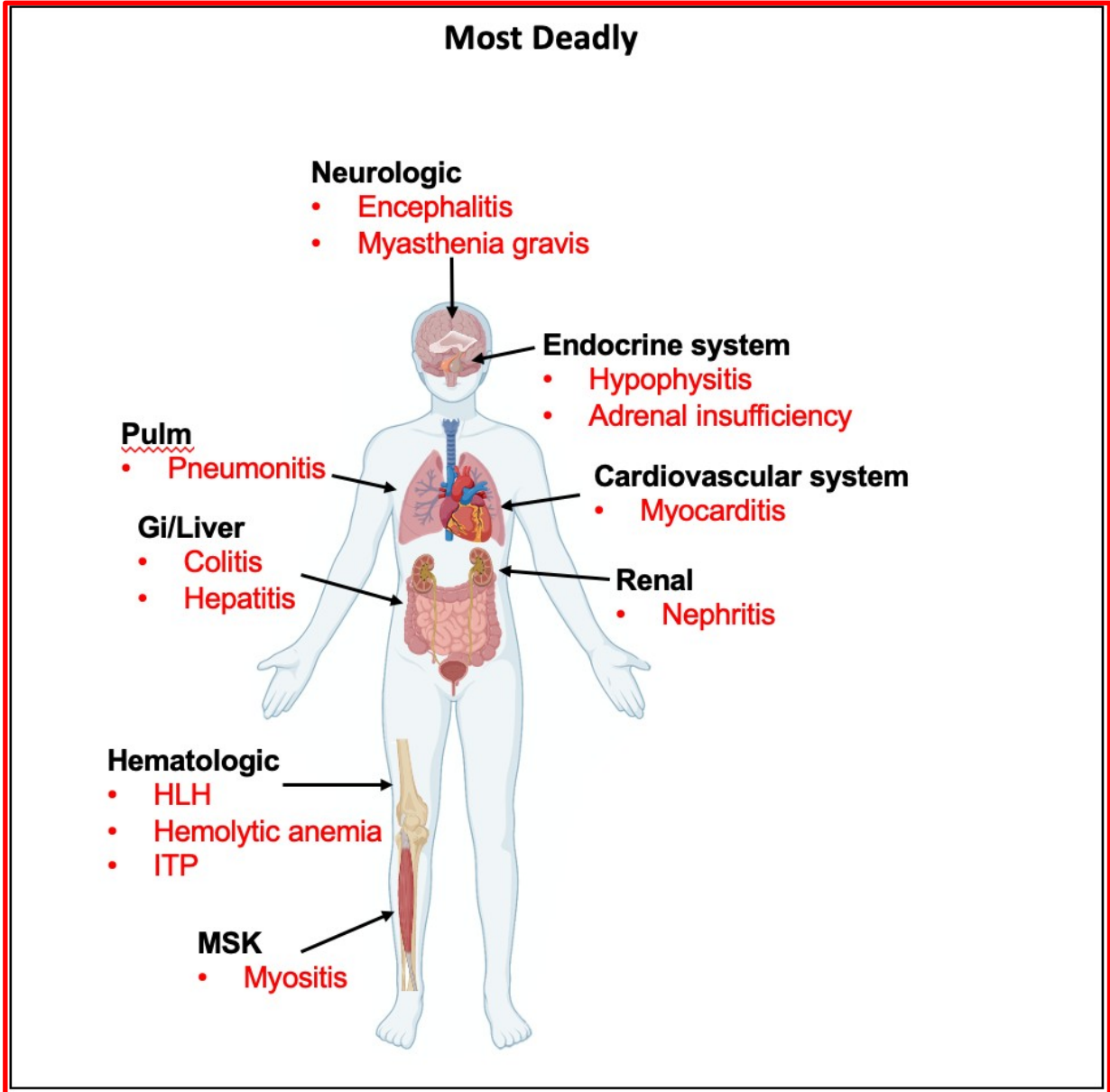
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All Side Effects Most common Irreversible Most Deadly



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All Side Effects Most common Irreversible **Most Deadly**



How do I treat immunotherapy toxicities?



Immunotherapy Toxicity Treatment Guidelines

Basic framework to treat ICI toxicity

- Rule out other causes
- Stop checkpoint inhibitor
- Grade severity
- Prednisone 1-2 mg/kg/day
- If not improving in a few days consider additional agents

Case 1

Case 1: A 46 yo woman w/hx of metastatic melanoma being treated with nivolumab (anti-PD-1) and ipilimumab (anti-CTLA-4) presents to your primary care office with new onset diarrhea. Having 5-6 BMs per day, not interfering with her ADLs/IADLs

What work up should you do?

- Check stool PCR (r/o C diff and other infectious causes)
- Lactoferrin/calprotectin

What grade colitis does she have?

- Grade 2

Infectious w/up comes back negative. What treatment should you offer?

- Hold nivolumab and ipilimumab
- Prednisone 1-2 mg/kg/day
- If no response in 2-3 days, continue steroids and consider adding infliximab or vedolizumab within 2 weeks

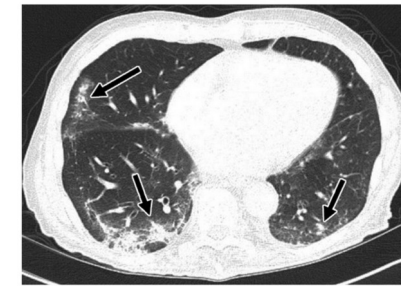
Case 2

Case 2: A 68 yo gentleman with stage IV NSCLC on single agent pembrolizumab (anti-PD-1). On his 5th cycle he develops a worsening cough and dyspnea on exertion and comes into your office to be examined found to have SpO2 of 88% on RA with coarse b/l crackles.

What initial work up should you do?

- Nasal swab to eval for viral infections
- Sputum culture
- CT-chest with contrast
- Consider pulm consult

CT-scan looks like this...



multifocal peripheral and subpleural mid- and lower-lung airspace consolidations (arrows)⁶

What treatment should you offer?

- Hold pembrolizumab
- Prednisone 1-2 mg/kg/day
- Consider abx if infection not ruled out
- If no response in 48 hrs, consider infliximab, IVIG or mycophenylate

2022–25 Update — New ICI Classes & irAE Guidance

New checkpoint target (3rd class): LAG-3.

Relatlimab+nivolumab (Opdualag) — first LAG-3-blocking combo; FDA approved Mar 2022 for unresectable/metastatic melanoma (RELATIVITY-047: median PFS 10.1 vs 4.6 mo).

More agents since the deck (Nov 2021):

Tremelimumab (anti-CTLA-4) + durvalumab — FDA approved 2022 (unresectable HCC, metastatic NSCLC). Dostarlimab (anti-PD-1) — dMMR endometrial/solid tumors.

irAE management framework unchanged (ASCO/NCCN, current):

G1 — continue ICI + monitor (exceptions: neuro/some heme). G2 — hold ICI, prednisone ~1 mg/kg, resume when $\leq$ G1.

G3 — hold + high-dose steroid (prednisone/methylpred 1–2 mg/kg), taper $\geq$ 4–6 wk; add steroid-refractory agent (e.g. infliximab) if no response in 48–72 h.

G4 — permanently discontinue ICI (exception: endocrinopathies controlled on hormone replacement).

Irreversible/most-deadly irAEs to flag: myocarditis (~40–50% fatal), type 1 DM, primary adrenal insufficiency; check TSH/FT4 at baseline & q4–6 wk.

Take-Home Points

Tumors evade T cells by engaging immune checkpoints (PD-L1→PD-1, ligands→CTLA-4) that put the brakes on the immune response.

ICIs (anti-PD-1/PD-L1, anti-CTLA-4, and now anti-LAG-3) release those brakes; best in high-mutation-burden cancers (melanoma, lung, bladder).

irAEs are autoimmune, can hit any organ, and may appear during or months after therapy — keep ICIs on the differential.

Management framework: rule out other causes → stop ICI → grade severity → prednisone 1-2 mg/kg → add 2nd agent if no improvement in a few days.

Use the NCCN/ASCO irAE guidelines as your go-to resource at the bedside.

Attribution & Optimization Note

Adapted from TeachIM.org — “Immune Checkpoint Inhibitors” (Natalie Longino MD, Christopher Geiger MD, Erin Schenk MD; exec. ed. Brandon Fainstad MD). Published Nov 2021.

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Correction made (this copy): slide 6 — “Hodkin’s” → “Hodgkin’s” (FDA-approved-cancers list). All original animations/BioRender graphics preserved; no other slide content altered.

Added context (source-cited, 2022–25): LAG-3 inhibitor class (relatlimab/Opdualag, FDA Mar 2022); tremelimumab+durvalumab; dostarlimab; ASCO/NCCN irAE grading framework.

Companion file: LearnerHandout.pdf (cases). Educational use only; not medical advice — oncology therapy evolves, have content clinician-reviewed before teaching.